ECE 447: Data Analysis and Machine Learning for Engineers
Assignment 2

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Problem 1

1(a) Histograms of each feature

Construct and visualize a histogram of each features.

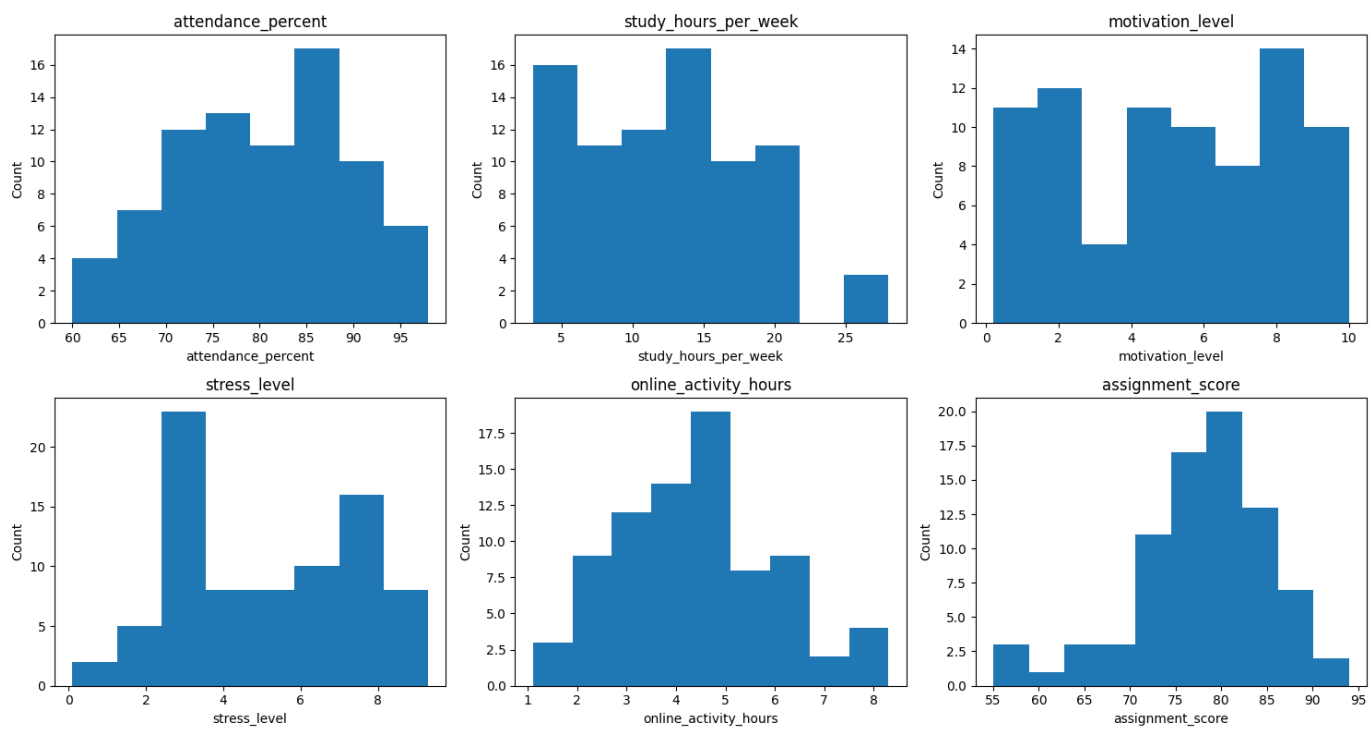

Figure 1: Histograms of all features (excluding identifier).

1(b) Mean, median, mode on histograms

Add three vertical lines on the histograms for mean, median, and mode of the distributions. Include proper legend for the figures.

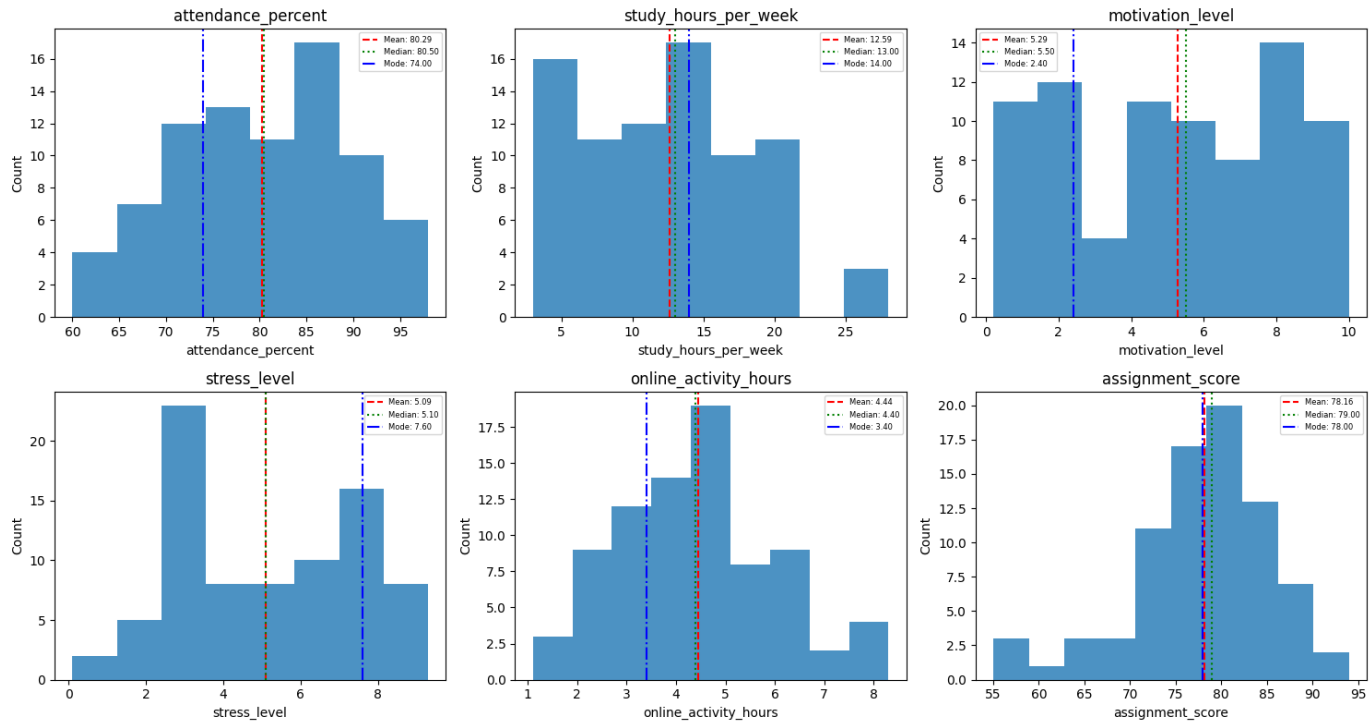

Figure 2: Histograms with mean, median, and mode overlays.

1(c) Distribution shape discussion

Report/discuss which distributions each feature resemble.
From the histograms, the features resemble the following distributions:

- **attendance_percent:** roughly unimodal and approximately normal, centered in the low–mid 80s with only mild skew.
- **study_hours_per_week:** right-skewed; most students cluster around typical study hours with a tail toward higher hours.
- **motivation_level:** bounded between 0 and 10 and appears irregular / multi-modal and not Gaussian.
- **stress_level:** bimodal; two noticeable clusters (lower stress and higher stress groups).
- **online_activity_hours:** roughly unimodal and close to normal around the middle of the range, with mild tails.
- **assignment_score:** roughly unimodal and near-normal around the high 70s, with a slight left tail due to a few low-score observations.

1(d) Scatter plots: assignment score vs other features

Visualize the following figures: y-axis to be assignment score, and x-axis to be rest of the features. Plot each in a separate figure.

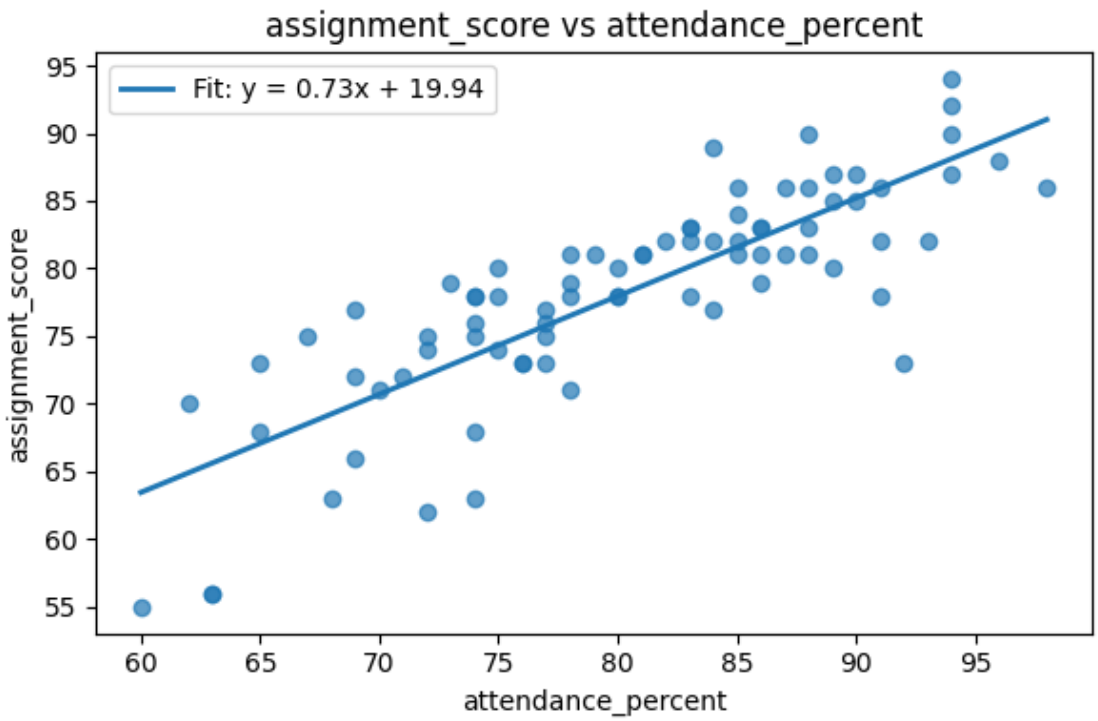

Figure 3: Assignment score vs attendance percentage.

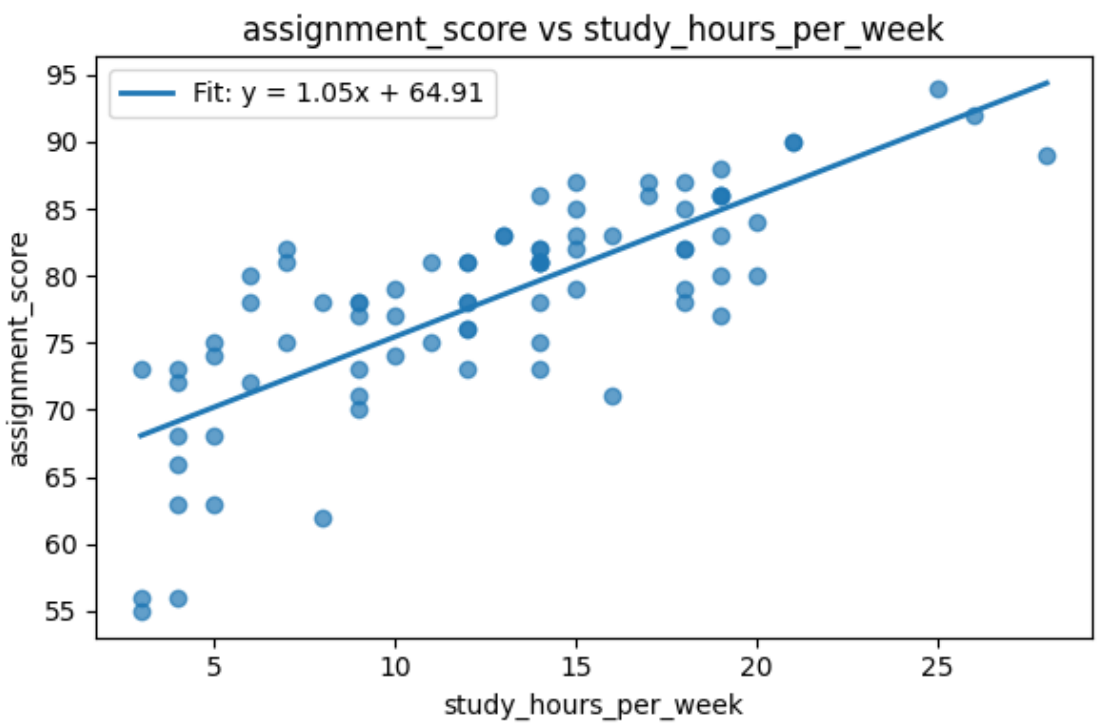

Figure 4: Assignment score vs study hours per week.

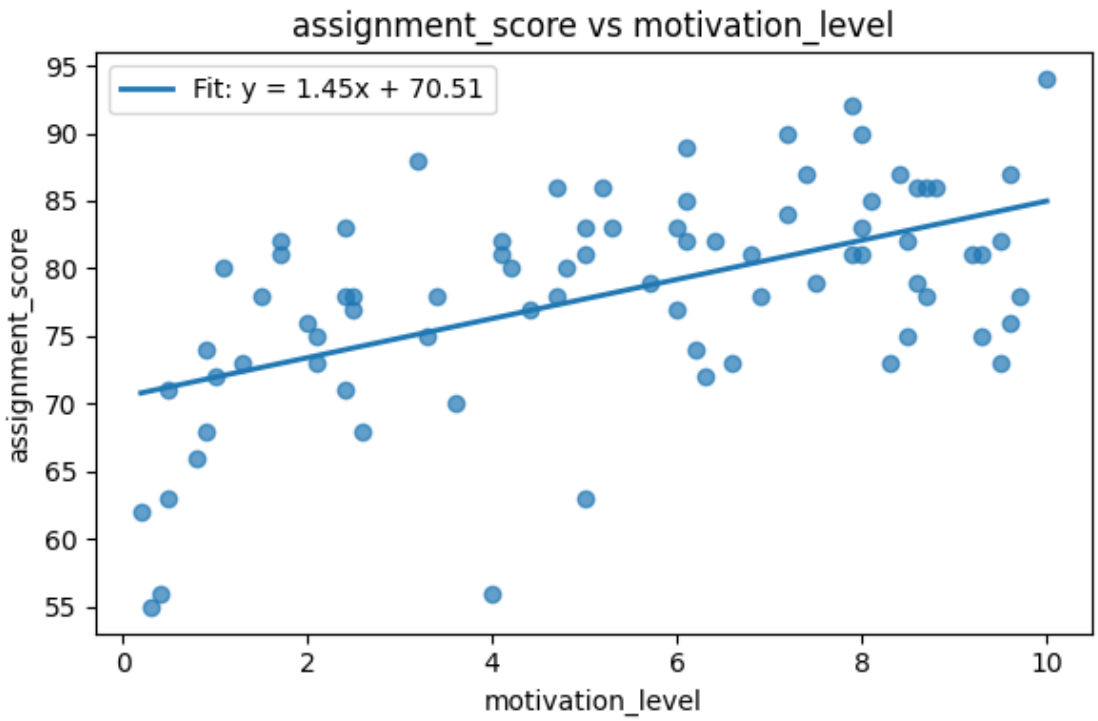

Figure 5: Assignment score vs motivation level.

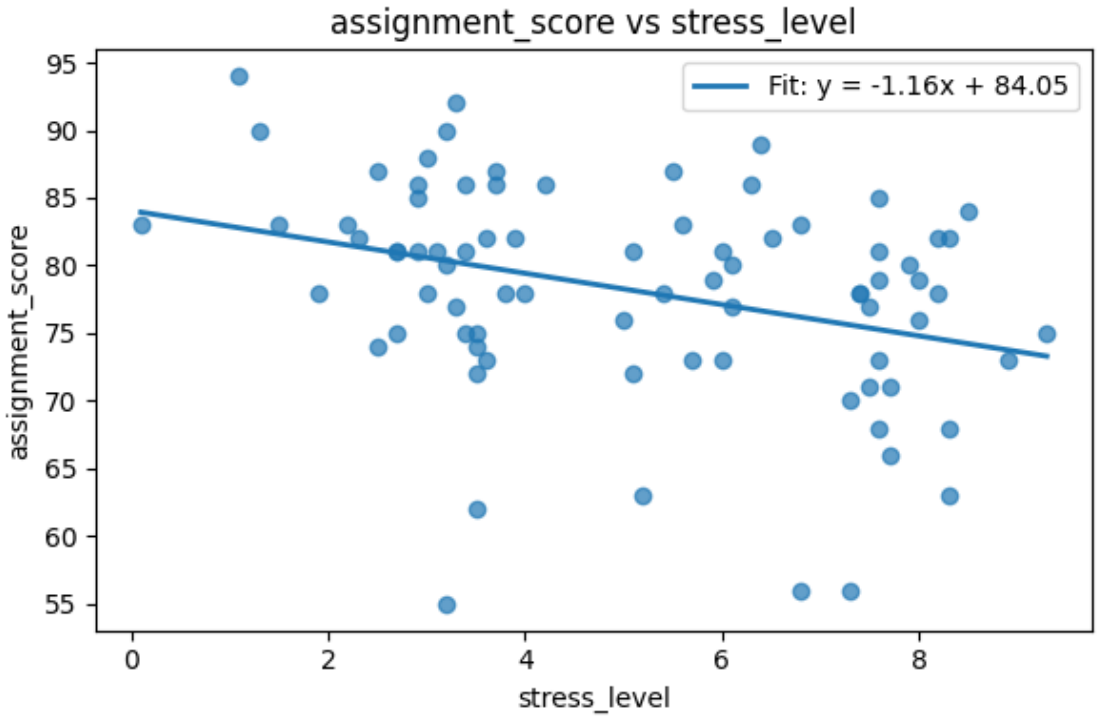

Figure 6: Assignment score vs stress level.

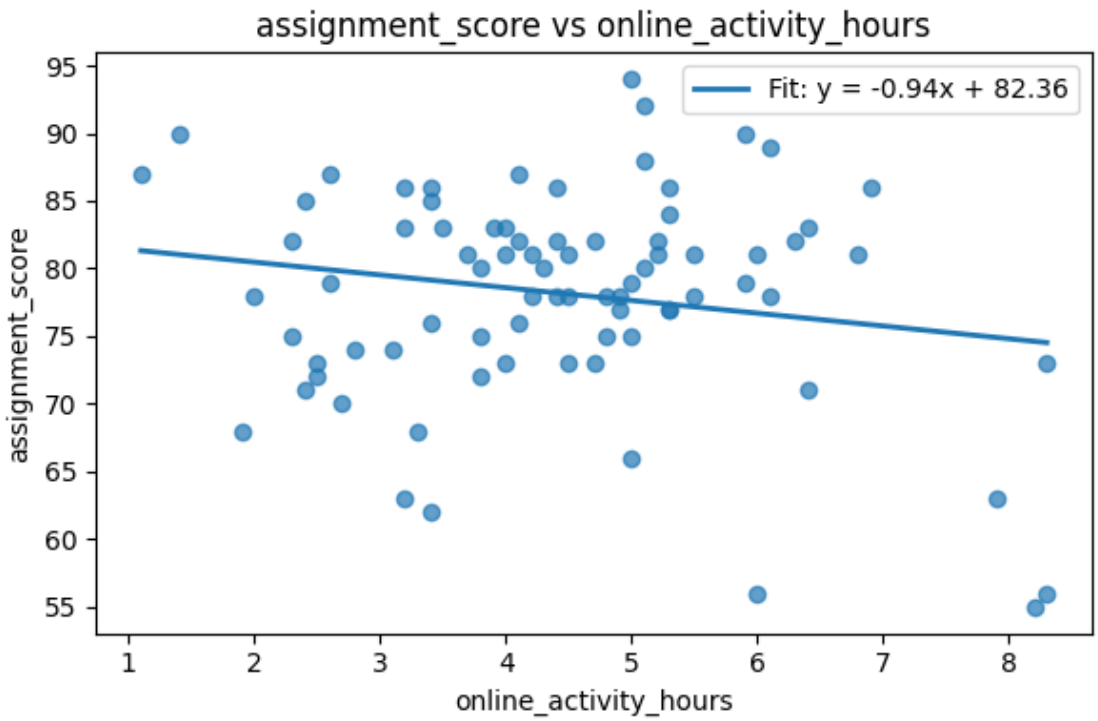

Figure 7: Assignment score vs online activity hours.

1(e) Correlation with assignment score

Calculate the correlation between assignment score and all other features.

Table 1: Correlation between assignment score and other features.

Feature	Correlation with assignment_score
attendance_percent	**0.824068**
study_hours_per_week	**0.779994**
motivation_level	**0.545948**
stress_level	**-0.334311**
online_activity_hours	**-0.184641**

1(f) Discussion of relationships

Discuss the relationships and comment on the obtained results.

The correlation results align with the scatter plots. `attendance_percent` and `study_hours_per_week` show strong positive relationships with `assignment_score` ($r = 0.824$ and $r = 0.780$), which matches the tight upward trends in their best-fit lines. `motivation_level` has a moderate positive correlation ($r = 0.546$), indicating motivation is associated with higher scores but with noticeably more variability than attendance or study time. In contrast, `stress_level` shows a weak-to-moderate negative relationship ($r = -0.334$), suggesting higher stress tends to correspond to lower scores, though the spread indicates it is not strongly predictive on its own. `online_activity_hours` has only a weak negative correlation ($r = -0.185$), consistent with a shallow downward trend and

substantial scatter. Overall, attendance and study hours appear to be the strongest individual predictors of assignment performance in this dataset.

Problem 2

Goal: predict passing course (PASS if assignment score ≥ 65, FAIL otherwise). Positive class is FAIL. Data: `data-a2-p2.csv`.

2(a) Confusion matrices (threshold = 0.50)

Using a classification threshold of 0.50 on the prediction score, construct the confusion matrix for each of the models. Use this threshold for b) and c) below.

Model 1

Table 2: Confusion matrix (Positive = FAIL) for Model 1 at threshold 0.50.

	Predicted FAIL (P)	Predicted PASS (N)
Actual FAIL (P)	2	18
Actual PASS (N)	68	12

Model 2

Table 3: Confusion matrix (Positive = FAIL) for Model 2 at threshold 0.50.

	Predicted FAIL (P)	Predicted PASS (N)
Actual FAIL (P)	5	15
Actual PASS (N)	80	0

2(b) Accuracy, ACA, harmonic-mean ACA

Calculate the simple (overall) accuracy, average class accuracy (ACA) and the average class accuracy with harmonic mean for each model.

Model 1: From the confusion matrix (TP=2, FN=18, FP=68, TN=12; total=100),

$$\text{Acc} = \frac{TP + TN}{100} = \frac{2 + 12}{100} = 0.14.$$

Class accuracies (FAIL is positive) are

$$\text{Acc}_{FAIL} = \frac{TP}{TP + FN} = \frac{2}{20} = 0.10, \quad \text{Acc}_{PASS} = \frac{TN}{TN + FP} = \frac{12}{80} = 0.15.$$

Thus,

$$ACA = \frac{0.10 + 0.15}{2} = 0.125, \quad ACA_{harm} = \frac{2(0.10)(0.15)}{0.10 + 0.15} = 0.12.$$

Model 2: From the confusion matrix (TP=5, FN=15, FP=80, TN=0; total=100),

$$\text{Acc} = \frac{TP + TN}{100} = \frac{5 + 0}{100} = 0.05.$$

Class accuracies are

$$\text{Acc}_{FAIL} = \frac{5}{20} = 0.25, \quad \text{Acc}_{PASS} = \frac{0}{80} = 0.$$

Thus,

$$ACA = \frac{0.25 + 0}{2} = 0.125, \quad ACA_{harm} = 0.$$

2(c) Which model performs best (based on a and b)?

Based on a) and b), discuss which model appears to perform best at this task?
Based on the confusion matrices and metrics at threshold 0.50, Model 1 performs best overall. It achieves higher overall accuracy (0.14 vs 0.05) and a higher harmonic-mean ACA (0.12 vs 0.00), indicating more balanced performance across the FAIL and PASS classes. While both models have the same ACA (0.125), Model 2 performs poorly on the PASS class (TN=0), which leads to a harmonic-mean ACA of 0.

2(d) Precision and recall at threshold 0.50

Compute precision and recall for each model. Based on this, discuss which model is better to use to make sure that we miss the least number of students to be given a warning.
Using threshold 0.50 with FAIL as the positive class, precision and recall are:

Model 1: (TP=2, FP=68, FN=18)

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{2}{2 + 68} = \frac{2}{70} = 0.0286, \qquad \text{Recall} = \frac{TP}{TP + FN} = \frac{2}{2 + 18} = \frac{2}{20} = 0.10.$$

Model 2: (TP=5, FP=80, FN=15)

$$\text{Precision} = \frac{5}{5 + 80} = \frac{5}{85} = 0.0588, \qquad \text{Recall} = \frac{5}{5 + 15} = \frac{5}{20} = 0.25.$$

To miss the least number of failing students (minimize FN / maximize recall), **Model 2** is preferable since it achieves higher recall (0.25 vs 0.10) and fewer missed FAIL cases (FN=15 vs FN=18) at threshold 0.50.

2(e) Precision & recall curves vs threshold

Generate precision and recall for a range of threshold on the prediction score and plot them on one figure. Which model is preferable based on this curve?

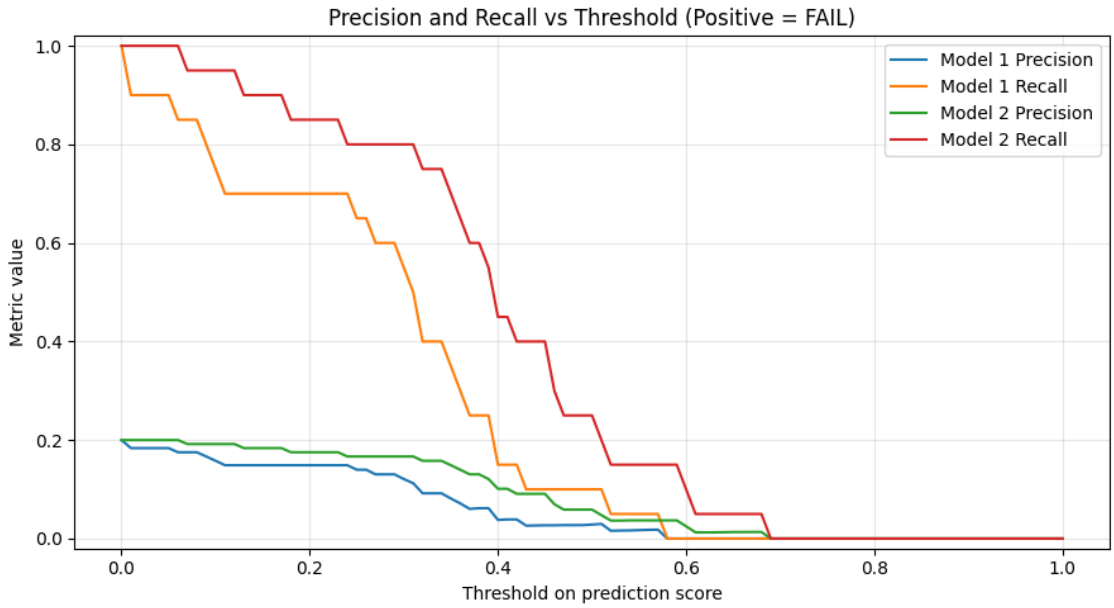

Figure 8: Precision and recall vs threshold for both models.

From the precision/recall vs threshold plot, **Model 2** is preferable when prioritizing detection of failing students (higher recall across most thresholds). However, both models have low precision overall, and Model 2 produces many false warnings (high FP) over a wide threshold range. Therefore, Model 2 is preferable for maximizing recall, while Model 1 may be preferable when the goal is to reduce false positives.

2(f) Threshold choice criteria

Explain how you may choose the threshold on the prediction score if you plan to minimize the warning given wrongly to passing students. Repeat for when you want to make sure you won't miss a warning for failing student. Which model do you suggest to use based on these criteria?

Based on the curves, **Model 2** is preferable when the main objective is to avoid missing failing students (higher recall). However, when the objective is to minimize wrong warnings to passing students (reduce false positives), **Model 1** is preferable because its false-positive rate decreases as the threshold increases, whereas Model 2 continues to warn nearly all passing students over a wide range of thresholds.

To make sure we do not miss warnings for failing students (false negatives), we should use a **lower threshold** to increase recall. In practice, the threshold can be decreased until recall reaches a target value (e.g., 0.90 recall), acknowledging that this will increase the number of false positives (more warnings to passing students).

Based on the precision/recall curves, **Model 2** is suggested under both criteria: it maintains higher recall across most thresholds (missing fewer failing students) and also shows similar or slightly higher precision compared to Model 1 over much of the threshold range.

2(g) Suggested threshold overall

Given the generated figures so far, discuss which threshold value on the prediction score you would suggest to use for identifying those with failing potentials, and give the least number of wrong warnings to already passing students.

To choose a threshold that identifies failing potential while minimizing wrong warnings to passing students (false positives), we compare FP/TN and recall across thresholds. Model 2 is not suitable for this objective in the tested range because it produces FP=80 and TN=0 for thresholds 0.30–0.50, meaning it warns essentially all passing students regardless of threshold. In contrast, Model 1 reduces false warnings as the threshold increases (e.g., FP decreases from 80 at $t = 0.30$ to 79 at $t = 0.35$, and TN increases from 0 to 1), but recall decreases.

A reasonable trade-off is **Model 1 with threshold** $t \approx 0.35$, which maintains a meaningful recall for failing students (recall = 0.40) while beginning to reduce wrong warnings to passing students (FP = 79, TN = 1). Higher thresholds further reduce false positives (e.g., FP = 68 at $t = 0.50$) but recall drops to 0.10, missing most failing students.

2(h) Cumulative gain

Construct (plot) the cumulative gain. Which model is better?

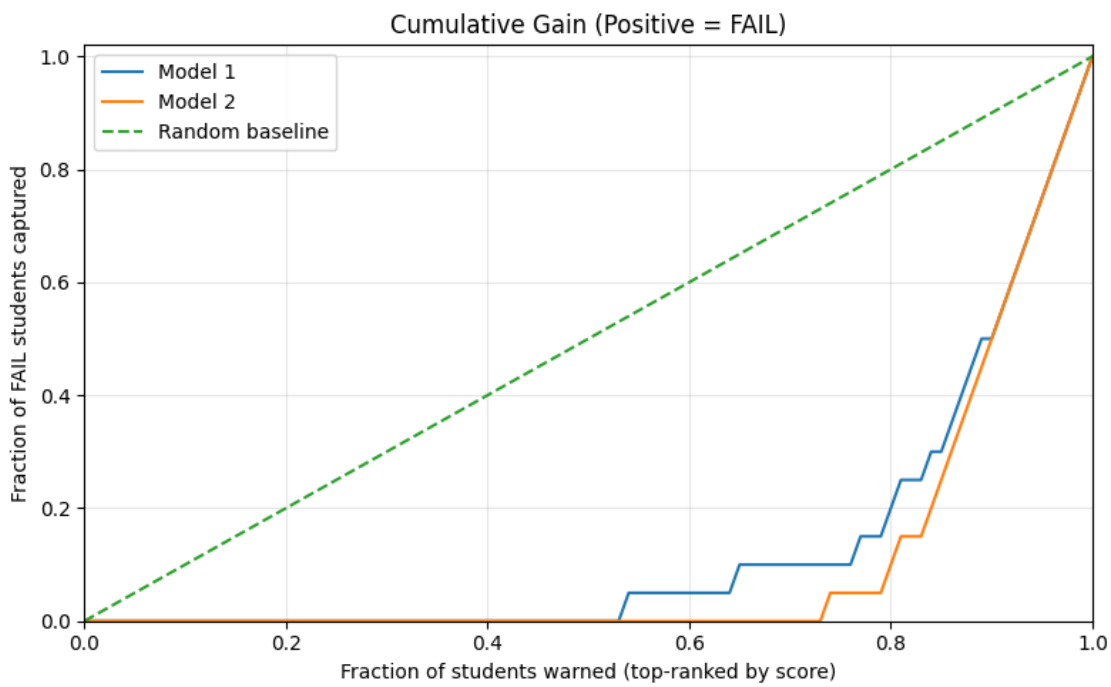

Figure 9: Cumulative gain curves for both models.

From the cumulative gain plot (positive = FAIL), both models perform poorly relative to the random baseline: for a large portion of the ranking, the fraction of failing students captured remains close to zero, meaning many students would need to be warned before capturing a substantial number of FAIL cases. Comparing the two models, **Model 1** generally achieves a higher cumulative gain than **Model 2** over most of the curve (i.e., for the same fraction of students warned, Model

1 captures a larger fraction of failing students). Therefore, based on cumulative gain, **Model 1 is the better model**, although neither model provides strong ranking performance.

2(i) ROC curves

Generate ROC curves for each model.

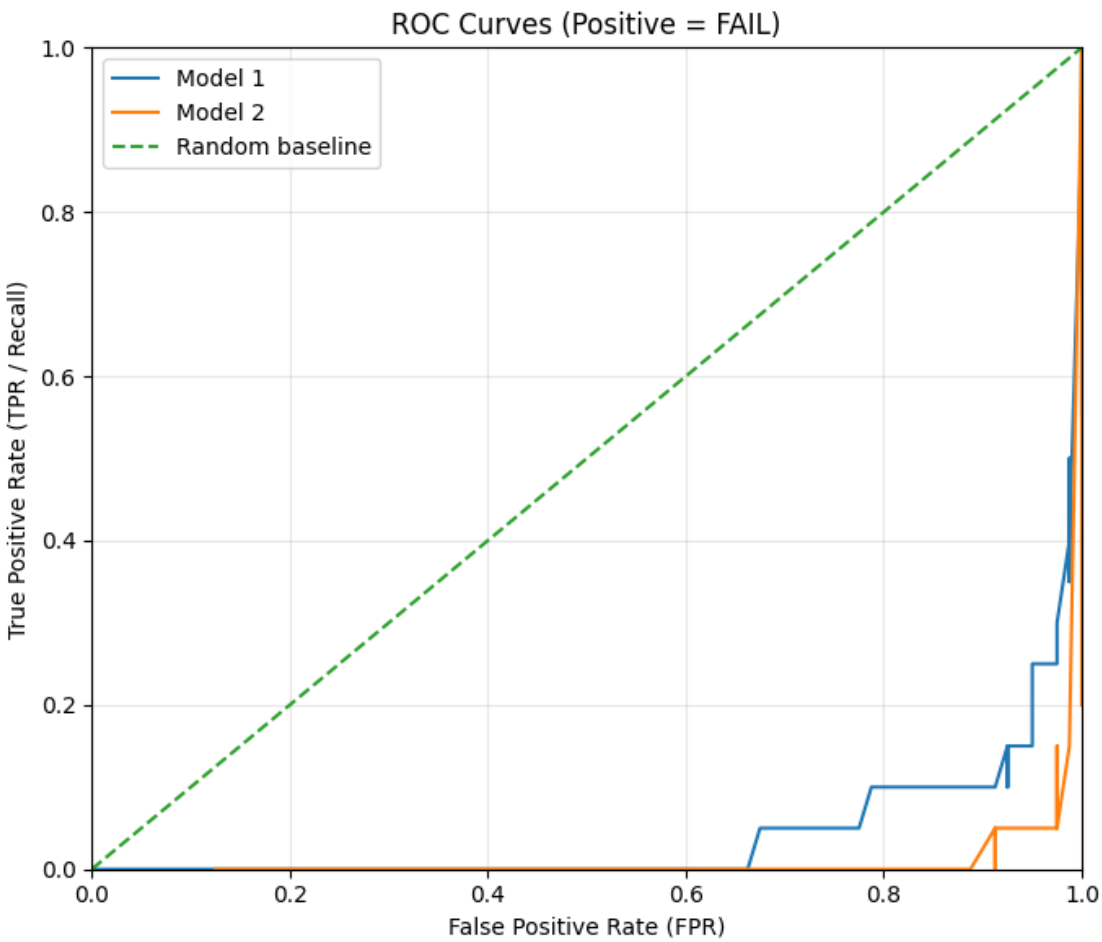

Figure 10: ROC curves for both models.

The ROC curves (positive = FAIL) indicate that both models have weak discriminative ability. The curves remain near a true positive rate close to zero until the false positive rate becomes very large, meaning the models must incorrectly flag many passing students in order to capture failing students. Comparing the two, **Model 1** generally achieves a higher TPR than **Model 2** for the same FPR (i.e., its ROC curve lies above Model 2 over most of the range). Therefore, based on the ROC curves, **Model 1 performs better**, although overall performance is poor.

2(j) Warn 90% of failing students

The instructor plans to warn 90% of the failing students. Based on the cumulative gain plot, would you recommend Model 1 or Model 2. Explain why and refer to some statistics from the cumulative gain.

To warn 90% of failing students, we use the cumulative gain curve to find the fraction of students that must be warned (x-axis) to capture 90% of FAIL cases (y-axis). Using the cumulative gain results, both models require warning approximately **98% of students** (fraction warned = 0.98) to reach 90% capture. This indicates that neither model ranks failing students near the top effectively, since nearly the entire class must be warned to catch most failing students. Based on this criterion alone, **Model 1 and Model 2 perform equivalently** for the 90% warning requirement (tie). If forced to choose, Model 1 can be slightly preferred because its cumulative gain curve is generally marginally higher over much of the range, but at the 90% target they are the same.